Devices to Place

Add these in Packet Tracer:

Device	Count
Router 1941	3
Switch 2960	3
PCs	3

Rename:

Device	Name
Router	R1
Router	R2
Router	R3
Switch	S1
Switch	S2
Switch	S3
PC	PC-A
PC	PC-B
PC	PC-C

---

2 Physical Connections

PC-A Network

```
PC-A Fa0 → S1 Fa0/2
S1 Fa0/1 → R1 Gig0/0
```

R1 to R2

```
R1 Se0/1/0 → R2 Se0/1/0
```

Use **Serial cable (DCE on one side)**.

R2 to R3

```
R2 Se0/1/1 → R3 Se0/1/0
```

PC-B Network

```
R2 Gig0/0 → S2 Fa0/1
S2 Fa0/2 → PC-B
```

PC-C Network

```
R3 Gig0/0 → S3 Fa0/1
S3 Fa0/2 → PC-C
```

---

3 Configure PC IP Addresses

PC-A

```
IP: 192.168.1.3
Mask: 255.255.255.0
Gateway: 192.168.1.1
```

PC-B

```
IP: 192.168.2.3
Mask: 255.255.255.0
Gateway: 192.168.2.1
```

PC-C

```
IP: 192.168.3.3
Mask: 255.255.255.0
Gateway: 192.168.3.1
```

---

4 Configure Router Interfaces

Configure R1

```
enable
configure terminal
```

LAN interface

```
interface g0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
```

WAN interface

```
interface s0/1/0
ip address 10.1.1.2 255.255.255.252
no shutdown
```

Configure R2

```
enable
conf t
interface g0/0
ip address 192.168.2.1 255.255.255.0
no shutdown
interface s0/1/0
ip address 10.1.1.1 255.255.255.252
no shutdown
interface s0/1/1
ip address 10.2.2.1 255.255.255.252
no shutdown
```

Configure R3

```
enable
conf t
interface g0/0
ip address 192.168.3.1 255.255.255.0
no shutdown
interface s0/1/0
ip address 10.2.2.2 255.255.255.252
no shutdown
```

---

5 Configure Static Routing

R1

```
ip route 192.168.3.0 255.255.255.0 10.1.1.1
```

R2

```
ip route 192.168.1.0 255.255.255.0 10.1.1.2
ip route 192.168.3.0 255.255.255.0 10.2.2.2
```

R3

```
ip route 192.168.1.0 255.255.255.0 10.2.2.1
```

---

6 Configure IPsec VPN on R1

Enter configuration mode.

```
enable
conf t
```

Step 1 — Define Interesting Traffic

```
access-list 100 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255
```

Step 2 — ISAKMP Policy (Phase 1)

```
crypto isakmp policy 10
encryption aes
hash sha
authentication pre-share
group 2
lifetime 86400
```

Step 3 — Pre-Shared Key

```
crypto isakmp key vpn123 address 10.2.2.2
```

Step 4 — Transform Set (Phase 2)

```
crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
```

Step 5 — Create Crypto Map

```
crypto map VPN-MAP 10 ipsec-isakmp
set peer 10.2.2.2
set transform-set VPN-SET
match address 100
```

Step 6 — Apply Crypto Map

```
interface s0/1/0
crypto map VPN-MAP
```

---

7 Configure IPsec VPN on R3

Enter CLI

```
enable
conf t
```

Interesting Traffic

```
access-list 100 permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255
```

ISAKMP Policy

```
crypto isakmp policy 10
encryption aes
hash sha
authentication pre-share
group 2
lifetime 86400
```

Pre-Shared Key

```
crypto isakmp key vpn123 address 10.1.1.2
```

Transform Set

```
crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
```

Crypto Map

```
crypto map VPN-MAP 10 ipsec-isakmp
set peer 10.1.1.2
set transform-set VPN-SET
match address 100
```

Apply Crypto Map

```
interface s0/1/0
crypto map VPN-MAP
```

---

8 Verify VPN

From **PC-A**

```
ping 192.168.3.3
```

This triggers the VPN tunnel.

---

9 Check VPN Status

On R1 or R3 run:

```
show crypto isakmp sa
```

and

```
show crypto ipsec sa
```

If VPN works you will see **QM_IDLE state**.

---

10 Expected Result

Traffic between:

```
192.168.1.0 network
AND
192.168.3.0 network
```

is **encrypted through the IPsec tunnel**.

Likely Viva Questions

What is VPN?
VPN (Virtual Private Network) creates a **secure encrypted tunnel over public networks**.

What is IPsec?
IPsec is a **protocol suite that secures IP communication using encryption and authentication**.

What is ISAKMP?
ISAKMP (Internet Security Association and Key Management Protocol) negotiates **security parameters for IPsec**.

What is Transform Set?
It defines **encryption and hashing algorithms used in IPsec tunnel**.